Docker - Data persistence

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm --name my_ubuntu -it ubuntu:latest
root@01d07c3cca2f:/# echo "hello world from Docker" > /home/test.txt
root@01d07c3cca2f:/# cat /home/test.txt
hello world from Docker
root@01d07c3cca2f:/# exit
exit
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm --name my_ubuntu -it ubuntu:latest
root@933de6803509:/# cat /home/test.txt
cat: /home/test.txt: No such file or directory
root@933de6803509:/# exit
exit
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker run --name my_time_keeper datascientest/time-keeper:latest
docker: Error response from daemon: Conflict. The container name "/my_time_keeper" is already in use by container "99631cd84657075d474f6c2d523703c408e71
6dbf5184e9812fcf9631dcd9795". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker start my_time_keeper
my_time_keeper
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker volume create my_volume
my_volume
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker volume ls
DRIVER    VOLUME NAME
local     7a9ec9b8008c728713733f6a09210135f458b321b5a9333adb4e466aee94b95b
local     14df03a573502f037f3913b13a4103a2f0d09df3c224f666c1e34cb0512bdf0d
local     19bf8064e5866bf21ac9dd5c7188e3e3db92ae3c1548fd3387c3694283d12f39
local     b5c91650e6c0dfc7895991759fda745b84577addf2c1b032c2014ef9e60fff4e
local     fe08dcf8fe0754a384d22cbf50424e7350dda3ae0c0fd44ecf286a01ff1ff062
local     my_volume
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker volume inspect my_volume
[
    {
        "CreatedAt": "2025-06-15T10:47:06Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/my_volume/_data",
        "Name": "my_volume",
        "Options": null,
        "Scope": "local"
    }
]
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it --name my_ubuntu \
>   --mount type=volume,src=my_volume,dst=/home/my_folder \
>   --rm ubuntu:latest bash
root@ca83fe111282:/# ls /home
my_folder  ubuntu
root@ca83fe111282:/# echo "hello world from Docker" > /home/my_folder/test.txt
root@ca83fe111282:/#
```

Switch to 2nd terminal to run:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ sudo ls /var/lib/docker/volumes/my_volume/_data
test.txt
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ sudo cat /var/lib/docker/volumes/my_volume/_data/test.txt
hello world from Docker
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ echo "hello world from the host machine" > ~/test2.txt
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ sudo mv ~/test2.txt /var/lib/docker/volumes/my_volume/_data/
test2.txt
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

Switch back to 1st terminal to run:

```
root@ca83fe111282:/# ls /home/my_folder
test.txt  test2.txt
root@ca83fe111282:/# cat /home/my_folder/test2.txt
hello world from the host machine
root@ca83fe111282:/# 
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it --name my_ubuntu \
>   --mount type=volume,src=my_volume,dst=/home/my_folder \
>   --rm ubuntu:latest bash
root@ca83fe111282:/# ls /home
my_folder  ubuntu
root@ca83fe111282:/# echo "hello world from Docker" > /home/my_folder/test.txt
root@ca83fe111282:/# ls /home/my_folder
test.txt  test2.txt
root@ca83fe111282:/# cat /home/my_folder/test2.txt
hello world from the host machine
root@ca83fe111282:/# 
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it \
> --name my_ubuntu1 \
> --mount type=volume,src=my_volume,dst=/home/my_folder1 \
> --rm ubuntu:latest bash
ls /home/my_folder1
root@dd2208d2067f:/# ls /home/my_folder1
test.txt  test2.txt
root@dd2208d2067f:/# 
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it \
>   --name my_ubuntu2 \
>   --mount type=volume,src=my_volume,dst=/home/my_folder2 \
>   --rm \
>   ubuntu:latest bash
root@0b5aff4bd03e:/# ls /home/my_folder2
test.txt  test2.txt
root@0b5aff4bd03e:/# cat /home/my_folder2/test.txt
hello world from Docker
root@0b5aff4bd03e:/# cat /home/my_folder2/test2.txt
hello world from the host machine
root@0b5aff4bd03e:/# 
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker volume rm my_volume
my_volume
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ 
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it\
>  --name my_ubuntu\
>  --volume $HOME:/home/my_folder\
>  --rm\
>  ubuntu:latest bash
root@85d493d87c53:/# ls /home/my_folder
Spoon-Knife  code1.py                        exam_DOHERTY.tar            github_actions   log           sales.txt
Test         code1_test.py                   exam_bashlinux_SIOBHAN      kennel_v1.sql    my_directory  script.sh
__pycache__  cront.txt                       exam_bashlinux_SIOBHAN.tar  kennel_v1.sql.1  my_file       script.sh.save
api          data.txt                        examen.sql                  kennel_v1.sql.2  name.py       test.sh
api.tar      datascientest-data-engineering  example.txt                 kennel_v2.sql    people.json   test_library.py
api.tar.1    directory2                      fic                         kennel_v3.sql    prenom        wallet.py
api.tar.2    errors_file                     file1                       kennel_v4.sql    repo          wallet_test.py
bashlinux    exam_DOHERTY                    file2                       library.py       root_content  webscraping.py
root@85d493d87c53:/# 
```